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Mandelbrot Upic

A Commodore 64 Ultimate demo that generates a Mandelbrot fractal on-device at 64 MHz turbo, packs it directly into Upic format (a 16-color, 384x256 border-color raster picture technique), displays it live as it renders, and lets you interactively pan around and zoom into any region of the result.

See CREDITS.md for full attribution.

Status: v1.0.0, feature-complete and confirmed working on real Ultimate 64 hardware.

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Controls

Generation starts automatically on launch and the picture builds up live, left to right. Once it completes, you're in **browse mode**:

Input	Action
W/A/S/D	Pan the current view (no-op at the default overview -- nothing to pan to)
Cursor keys (hold Shift for up/left)	Same, alternative for muscle memory
Z	Enter box mode to pick a zoom target
O	Zoom out one notch (widens the view, clamped to the original overview)
C	Cycle the base color gradient (blue/orange, fire, ice, rainbow)

Panning regenerates the fractal at the same zoom level, shifted -- each step is a fraction of the current view's own size, so it moves less in absolute terms the deeper you've zoomed in.

Press Z to enter **box mode**: 4 corner markers (solid white 2x2 blocks) appear, outlining a box that always keeps the picture's own 3:2 aspect ratio:

Input	Action
W/A/S/D	Move the whole box
Cursor keys (hold Shift for up/left)	Same, alternative for muscle memory
+	Grow the box (aspect ratio unchanged)
-	Shrink the box (aspect ratio unchanged)
RETURN	Confirm and zoom into the box
Z	Cancel back to browse mode without zooming
C / O	Same as browse mode

Confirming regenerates the fractal at the selected region and returns to browse mode. Repeated zooms compose relative to whatever's currently displayed, so zooming, panning, and zooming again all work together.

Zoom precision limit: the fractal coordinates use a fixed-point format with a finite number of fractional bits, capping how far repeated zooming can go (roughly 16x total from the initial view) before individual pixel steps round down to zero and the picture stops changing with further zoom. This is a hard limit of the current implementation, not a bug -- see docs/MANDELBROT_ALGORITHM.md.

No quit key -- reset or power off to exit, same as many C64 demos with no graceful exit path. See docs/ZOOM_FEATURE.md for why.

Installation

Requires an **Ultimate 64 / Ultimate 64 Elite 2, firmware 3.15 or newer**.

1. Copy `build/mandelupic.prg` (or extract the release ZIP, which places it at `idi8b/mandelupic/mandelupic`) onto your Ultimate's SD card or USB storage.
2. Load the provided config file (`config/MandelbrotUpic-U64E2.cfg`, for Ultimate 64 Elite 2 boards) via the Ultimate's own configurator (F2 menu → Configuration → Load from file) to enable the Command Interface (UCI) and U64 turbo registers this demo needs. If you already have Command Interface and U64 Turbo Registers enabled in your own configuration, this step isn't necessary.
3. Run `mandelupic.prg` from the Ultimate's file browser.

Building from source

Prerequisites

Tool	Purpose	Install
Oscar64	C cross-compiler targeting 6502/C64	build from source, see their README
wput	FTP deploy to the Ultimate device	<code>sudo apt install wput</code>
pandoc + texlive-xetex	Generate README.pdf (optional)	<code>sudo apt install pandoc texlive-xetex</code>

Deploy setup

Copy `.env.example` to `.env` and set `ULTIP1` to your Ultimate device's IP address:

```
cp .env.example .env
```

Make targets

Target	Effect
<code>make /make all</code>	Compile to <code>build/mandelupic.prg</code> , regenerate <code>README.pdf</code> , build the release ZIP
<code>make deploy</code>	FTP the compiled <code>.prg</code> to the Ultimate device set in <code>.env</code>
<code>make docs</code>	Regenerate <code>README.pdf</code> via <code>pandoc</code>
<code>make clean</code>	Remove build outputs

Documentation

Document	Covers
docs/ARCHITECTURE.md	Program flow, components, memory layout overview
docs/MANDELBROT_ALGORITHM.md	Fixed-point fractal generation algorithm
docs/UPIC_VIEWER.md	The Upic border-color raster display technique
docs/ZOOM_FEATURE.md	Interactive pan/zoom/palette control scheme
TURBOCONTROLMANUAL.md	Ultimate 64 CPU speed control library
UCILIBMANUAL.md	Ultimate Command Interface (UCI) protocol library
CHANGELOG.md	Version history
CREDITS.md	Full attribution